

Analysis Case Study

Absences at Work

Assignment 1

740
Employee Records

6.9
Avg Absence/Record

36
Number of Employees



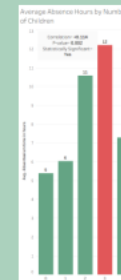
Who is more prone to absence?

Those with more children (2 & 3) had twice as many absentee hours as those with less children (0 & 1).

Social drinkers, on average, were absent 38% more than non-drinkers, while those with high school educations as their highest education were 35% more absent than those with masters, doctor, and postgraduate degrees.

Drinker	7.7	High School	7.2
Non-Drinker	5.9	Graduate	6.4
		Master / Doctor	5.3
		Postgraduate	5.3

Note: Individuals with multiple absences may have higher average absence hours than those with fewer absences. This is not necessarily a reflection of the individual's absence rate, but rather a reflection of the number of absences they have.



Distance (km) vs Absentee Time



Recommendations

- 1) Investigate "Sick" absenteeism:
 - The middle score group had the highest average absence hours, more than the extreme groups.
 - Disengagement can be a sign of disengagement, but more damaging to productivity.
 - Unhappy employees complain, which at least gives management a signal to act on. Disengaged employees are invisible. They show up late, leave early, take more sick days, and eventually leave without warning.
 - Re-engage this group with stay interviews to find hidden dissatisfaction.
- 2) Strengthen manager relationships:
 - Manager sentiment was the only positive dimension.
 - Invest in leadership training and build an army.
- 3) Act on retention risk:
 - 78% are unlikely to stay.
 - Focus on exit interview data, stay interviews, increasing compensation, employee growth paths.
- 4) Add more Dimensions:
 - Current 4 question model lacks specificity.
 - Add dimensions like pay, growth, workload, physical safety, etc.



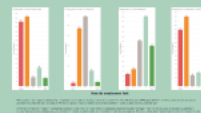
Job Satisfaction

Assignment 2

740
Employee Records

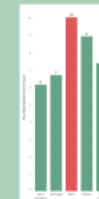
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Happiness Score vs Absenteeism

- 66% employees like their manager.
- 56% employees are unhappy.
- 78% employees unlikely to stay.
- "Well" workers averaged 18 hours absent, higher than unhappy (7 hours) and very unhappy (6 hours).



Consult Utilization

Assignment 3

740
Employee Records

6.9
Avg Absence/Record

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Absences at Work

Assignment 1

740

Employee Records

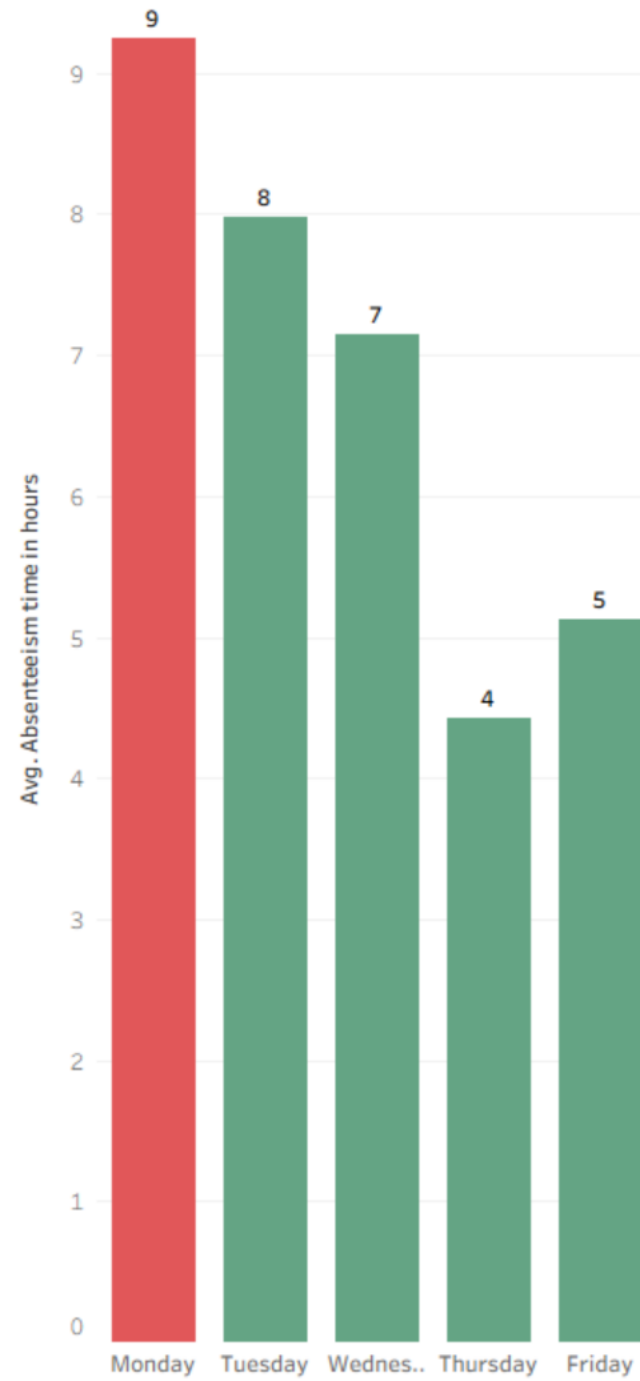
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Number of Employees

When do absences spike?



Monday had the highest average absentee time with 9 hours which was over 2 times higher than Thursday (4 hours).

July peaked at 9.1 hours while November and December also saw high peaks around 8 hours.



Who is more prone to absence?

Those with more children (2 & 3) had twice as many absentee hours as those with less children (0 & 1).

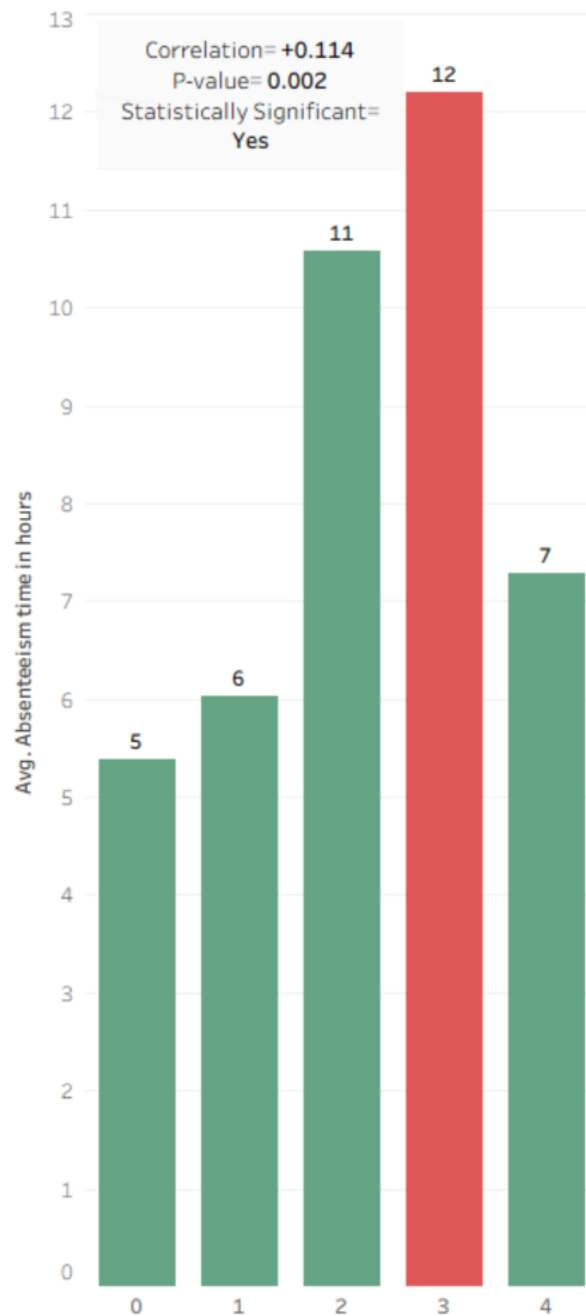
Social drinkers, on average, were absent 30% more than non-drinkers, while those with high school educations as their highest education were 35% more absent than those with masters, doctor, and postgraduate degrees.

Drinker	7.7
Non-Drinker	5.9

High School	7.2
Graduate	6.4
Master / Doctor	5.3
Postgraduate	5.3

Note: I only calculated correlations where the data was truly numeric: number of children and distance from work. For everything else I used averages, which are simpler and easier to understand.

Average Absence Hours by Number of Children



Distance (km) vs Absentee Time

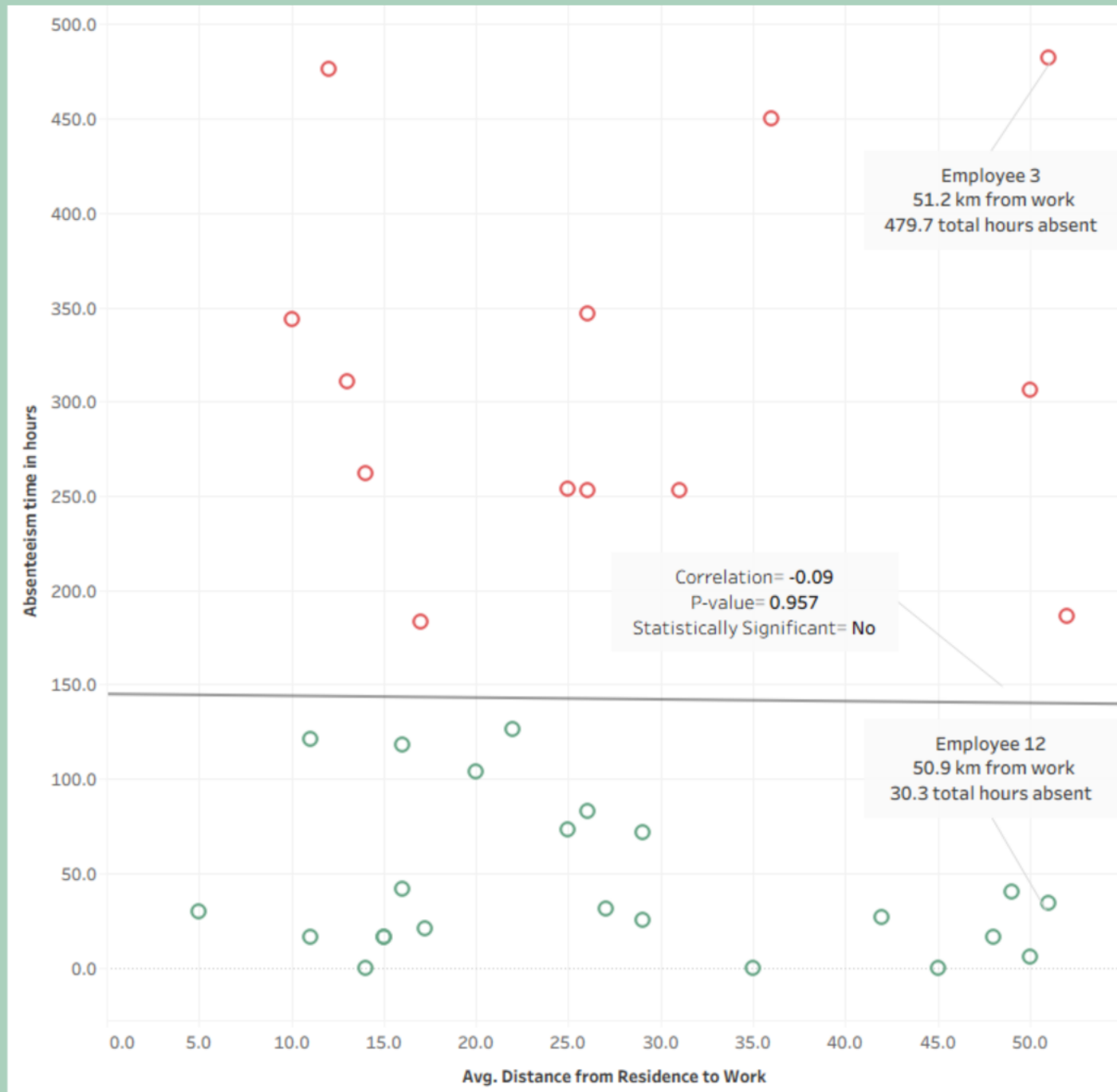
With 740 records correlation is -0.088 with a p-value of 0.016 which technically passes the significance threshold.

The 740-record correlation is misleading because the same 36 employees appear multiple times, they are not independent observations. 35/36 (97%) employees had the same distance repeated across all records. Employee 29 had two (15km, 26km).

Pearson correlation assumes each row is a separate, independent data point. Running it on repeated rows of the same people violates that assumption.

This means the result is not just small or unimportant it is invalid, produced by the wrong unit of analysis. Correcting for this by running the correlation on the 36 unique employees gives a p-value of 0.957 , confirming there was never a real relationship between distance and absence.

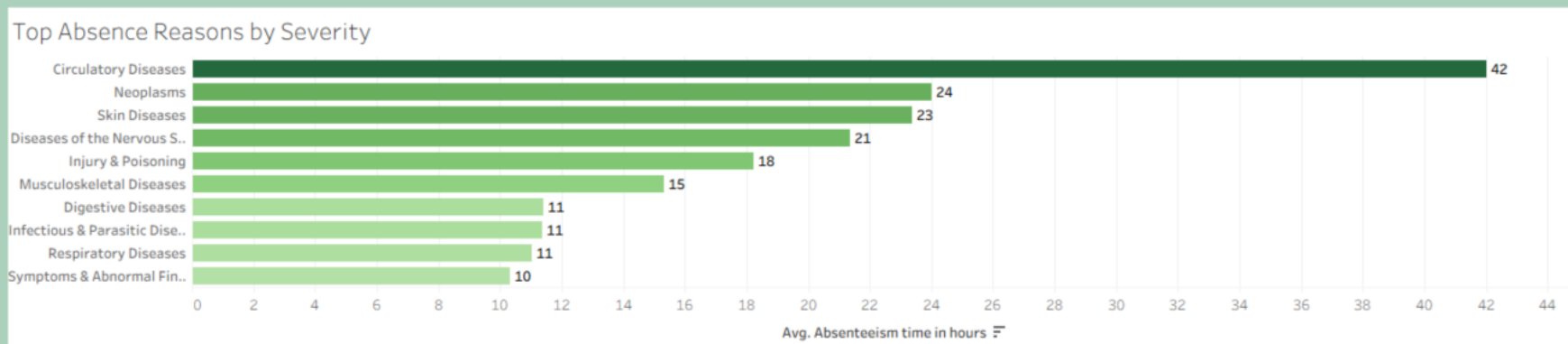
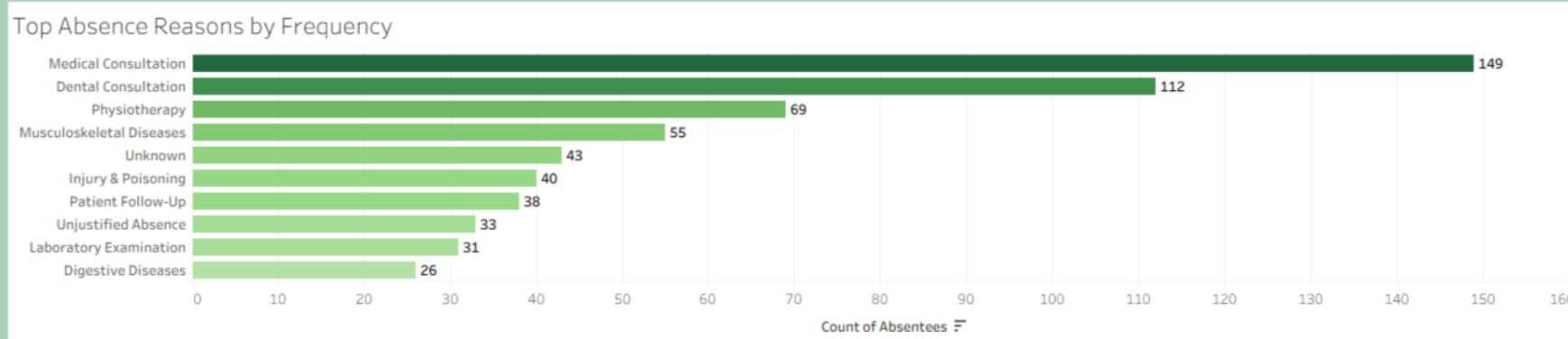
Employee 3 has 480 total absence hours.
Employee 12 has only 30 total absence hours.
A 16x difference despite identical commutes.



The Top Reason...

Recommendations

- 1) Manage routine medical absences:
 - Top 3 make up 330 absence records (44.5%).
 - **Encourage better scheduling, on-site/telehealth options.**
- 2) Target high severity cases:
 - Circulatory diseases, neoplasms, skin diseases make up top 3.
 - **Early intervention and wellness programs.**
- 3) Address Monday absenteeism directly:
 - Monday highest average absentee time.
 - **Flexible hours, cross-train to ensure coverage, less important meetings on Monday.**
- 4) Family support programs:
 - Families with 2+ children had more average absentee time.
 - **Childcare assistance, WFH options.**
- 5) Other wellness initiatives:
 - Social drinkers had a 30% higher absentee rate.
 - **Workplace wellness program for drinking.**



Job Satisfaction

Assignment 2

740

Employee Records

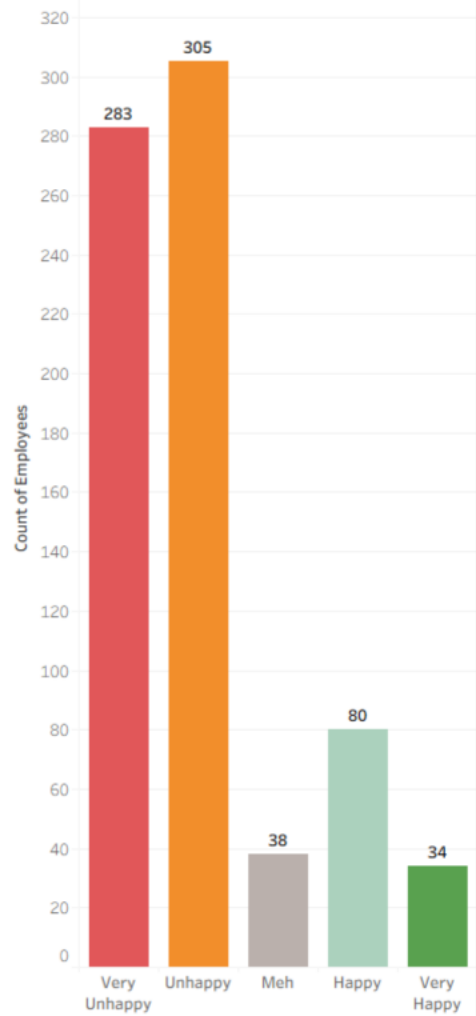
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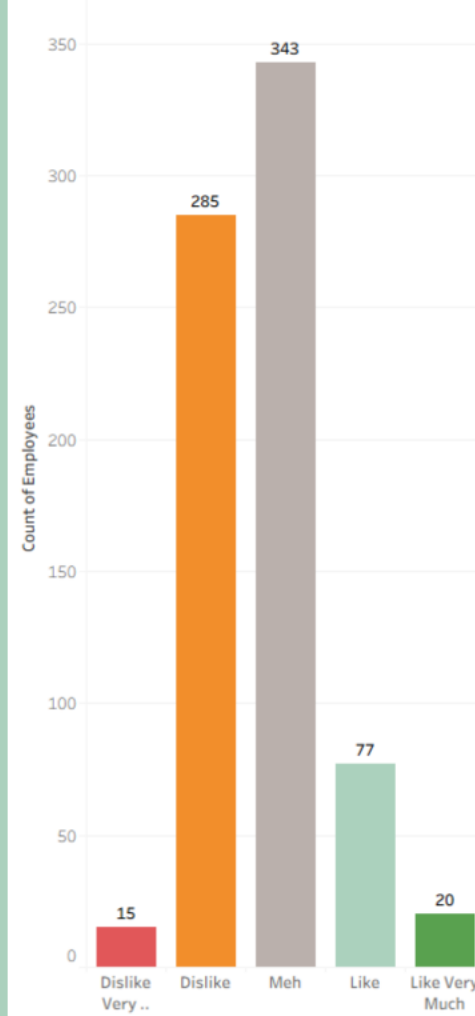
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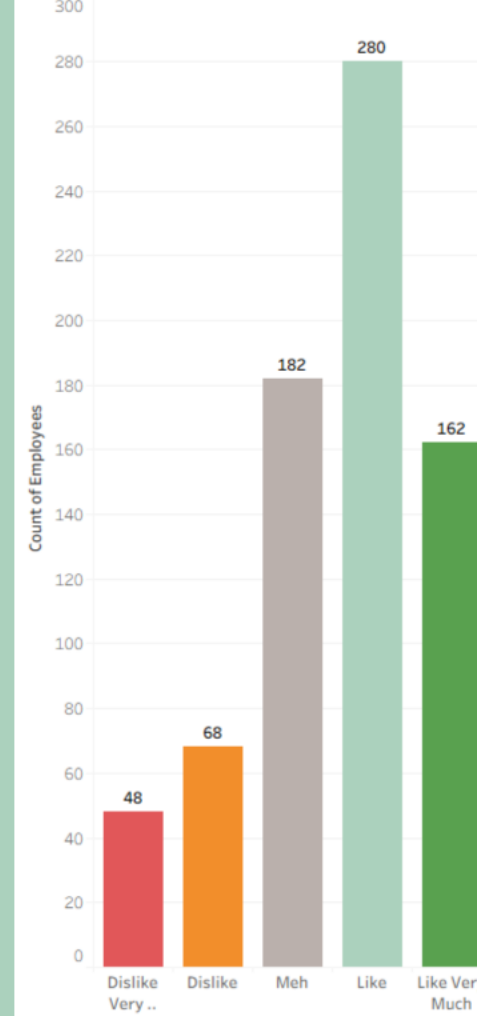
Employees vs Work Happiness



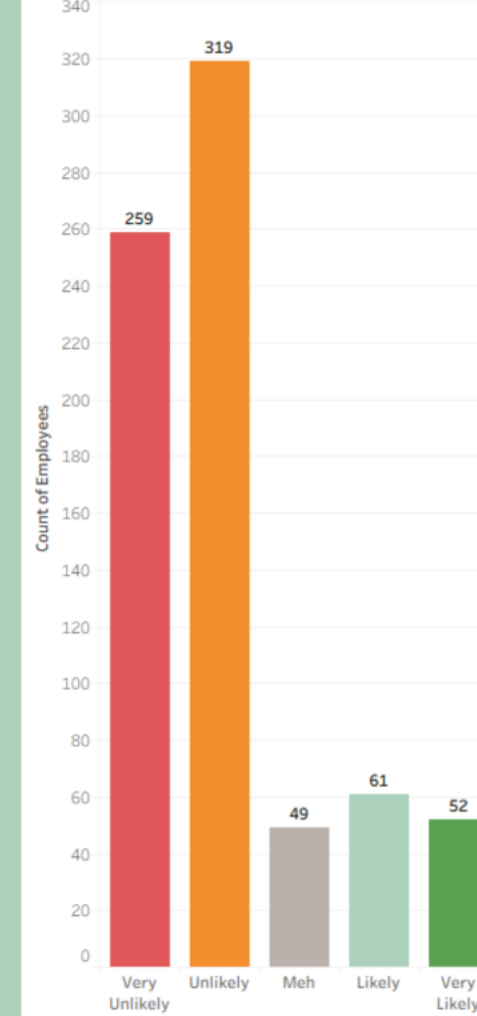
Employees vs Likes Colleagues



Employees vs Likes Manager



Employees vs Retention Outlook



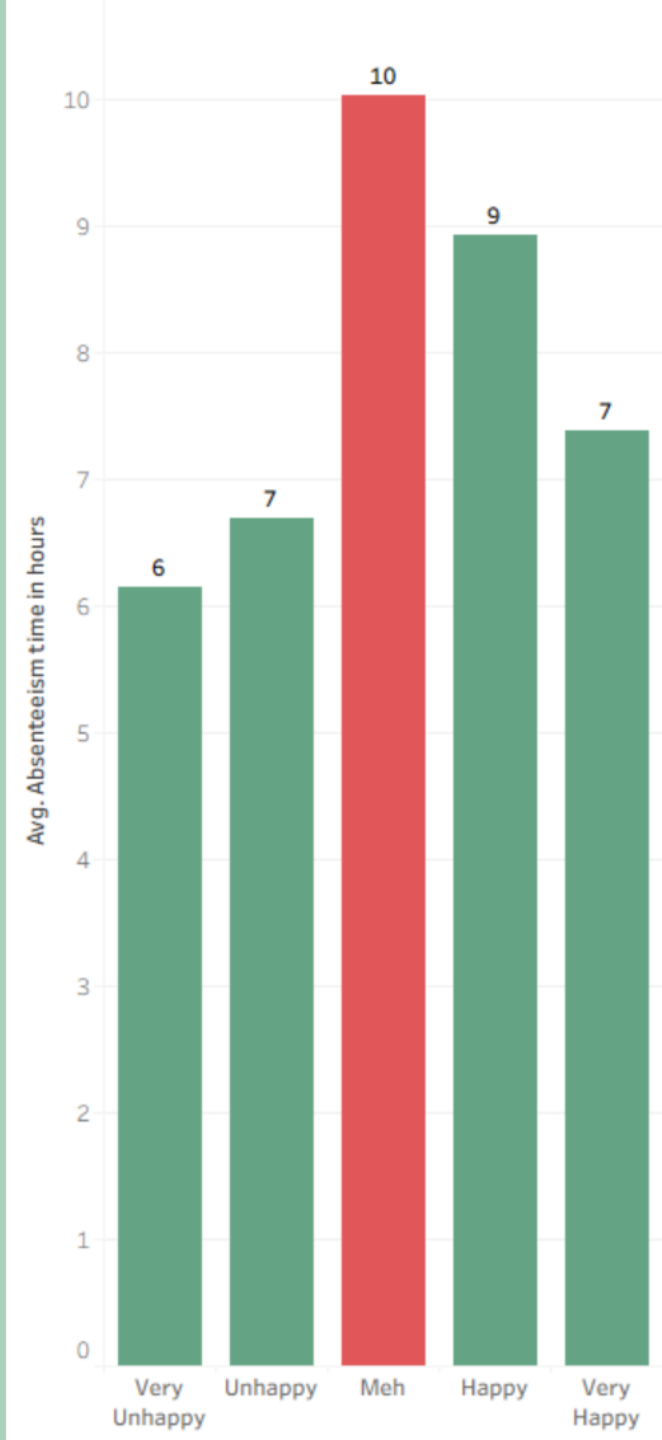
How do employees feel

740 records or 36 unique employees: Each employee has multiple survey responses that vary over time. 34 of 36 employees (94%) gave different answers across at least one survey question throughout the year, so using all 740 records gives a more complete picture of how sentiment relates to absence throughout the year.

Of the four dimensions, manager relationships stood out as the only bright spot. 60% of employees reported liking their manager, making it the only area where positive sentiment outweighed negative. Everything else, work satisfaction, colleague relationships, and retention outlook, skewed negative. This tells us that while management is a strength to build on, the broader work environment is driving disengagement.

Happiness Score vs Absenteeism

- 60% employees like their manager.
- 80% employees are unhappy.
- 78% employees unlikely to stay.
- "Meh" workers averaged 10 hours absent, higher than unhappy (7 hours) and very unhappy (6 hours).



Recommendations

1) Investigate "Meh" disengagement

- The middle score group had the highest average absence hours even more than the extreme groups.
- Disengagement can be quieter than unhappiness but more damaging in practice.
- Unhappy employees complain, which at least gives management a signal to act on. Disengaged employees are invisible: they show up late, leave early, take more sick days, and eventually leave without warning.
- **Re-engage this group with stay interviews to find hidden dissatisfaction.**

2) Strengthen manager relationships

- Manager sentiment was the only positive dimension.
- **Invest in leadership training and build on strength.**

3) Act on retention risk

- 78% are unlikely to stay.
- **Focus on exit interview data, stay interviews, increasing compensation, employee growing paths.**

4) Add more Dimensions

- Current 4 question model lacks specificity.
- **Add dimensions like pay, growth, workload, physical safety, etc.**

Consult Utilization

Assignment 3

16

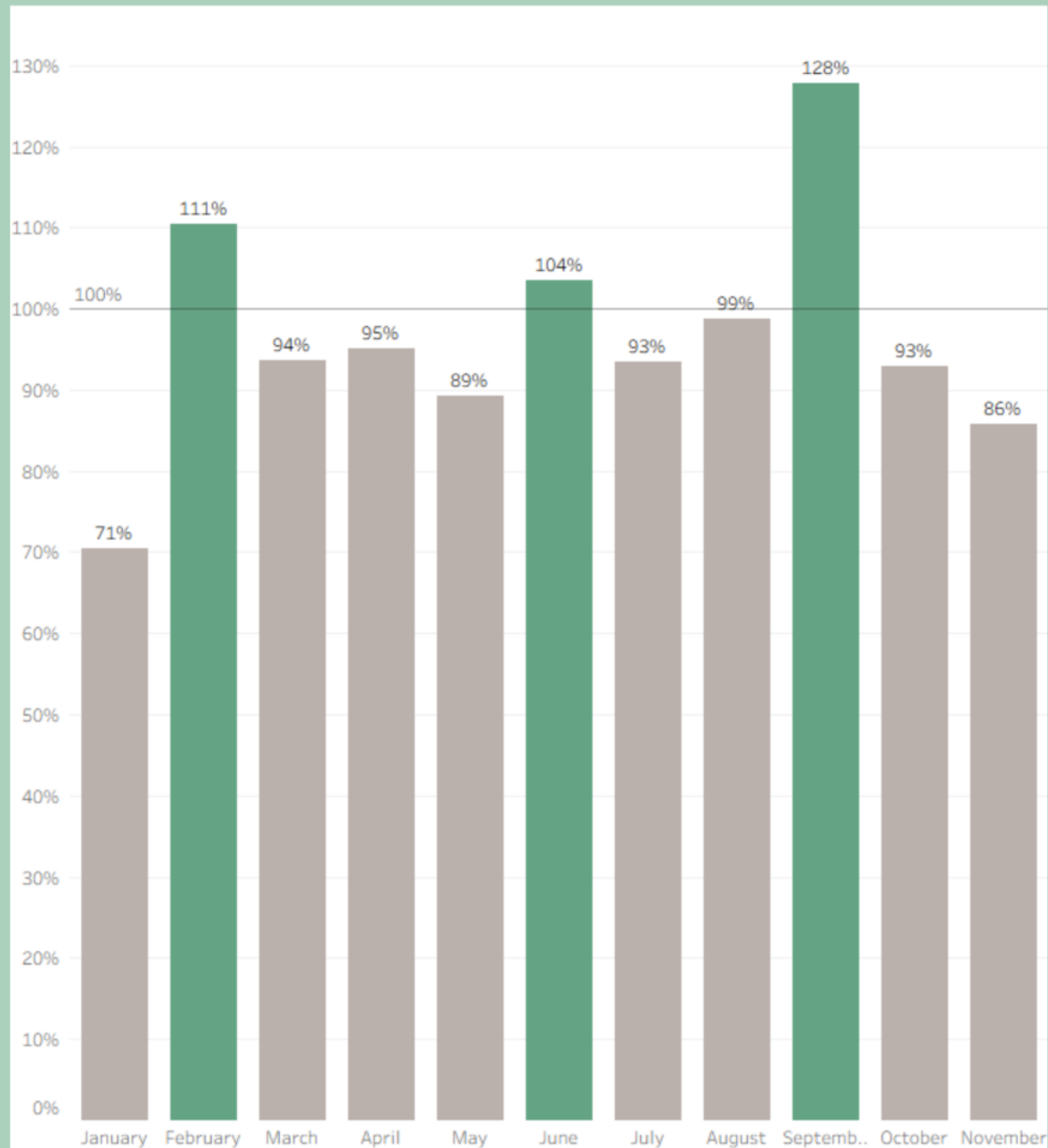
Number of Consultants

1,400

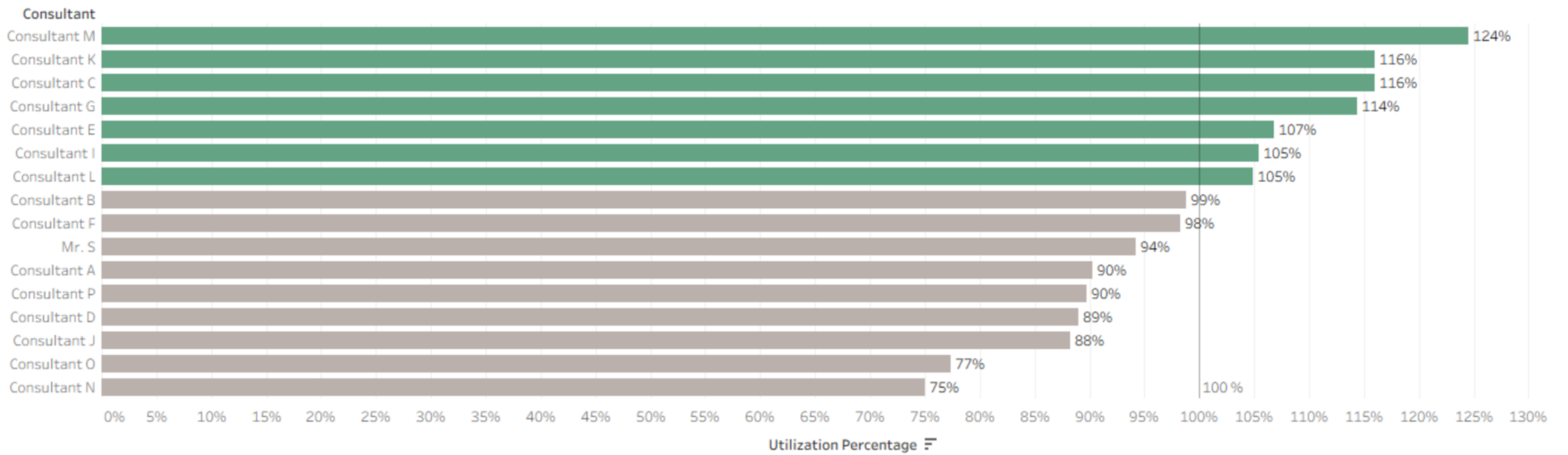
Annual Capacity Target

97%

Overall Utilization Rate



- Only 3 months hit the 100% target.
- September had the largest spike with 128%, while January had the lowest with 71%.
- The rolling averages of Jan-Mar was the lowest with 92%, while July-Sep had the highest with 107%.



7 of the consultants delivered above 100% utilization, with consultant M delivering the most at 124%.
9 of the consultants failed to reach 100%, with consultant N delivering the lowest at 75%.

Recommendations

- 1) Address underperforming consultants:
 - More than half of the consultants underperformed.
 - **Performance reviews, skill gaps, resolve planning issues.**
- 2) Investigate the spike in September:
 - Result of certain project?
 - **Plan workload distribution, prevent burnout.**
- 3) Review capacity planning:
 - Overall utilization rate good at 97% but the difference between highest month (September) and lowest month (January) was a 57% spread.
 - **More accurate capacity forecasting.**
- 4) Recognize top performers:
 - Consultant M, K, C, G overperformed.
 - **Recognize and retain, could provide model for the whole team.**

Analysis Case Study

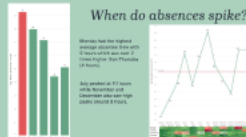
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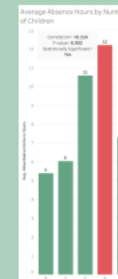
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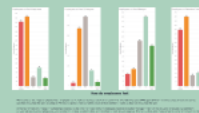
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